

Plozasiran (ARO-APOC3) for Severe Hypertriglyceridemia The SHASTA-2 Randomized Clinical Trial

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 Supplemental content

IMPORTANCE Severe hypertriglyceridemia (sHTG) confers increased risk of atherosclerotic cardiovascular disease (ASCVD), nonalcoholic steatohepatitis, and acute pancreatitis. Despite available treatments, persistent ASCVD and acute pancreatitis-associated morbidity from sHTG remains.

OBJECTIVE To determine the tolerability, efficacy, and dose of plozasiran, an *APOC3*-targeted small interfering-RNA (siRNA) drug, for lowering triglyceride and apolipoprotein C3 (APOC3, regulator of triglyceride metabolism) levels and evaluate its effects on other lipid parameters in patients with sHTG.

DESIGN, SETTING, AND PARTICIPANTS The Study to Evaluate ARO-APOC3 in Adults With Severe Hypertriglyceridemia (SHASTA-2) was a placebo-controlled, double-blind, dose-ranging, phase 2b randomized clinical trial enrolling adults with sHTG at 74 centers across the US, Europe, New Zealand, Australia, and Canada from May 31, 2021, to August 31, 2023. Eligible patients had fasting triglyceride levels in the range of 500 to 4000 mg/dL (to convert to millimoles per liter, multiply by 0.0113) while receiving stable lipid-lowering treatment.

INTERVENTIONS Participants received 2 subcutaneous doses of plozasiran (10, 25, or 50 mg) or matched placebo on day 1 and at week 12 and were followed up through week 48.

MAIN OUTCOMES AND MEASURES The primary end point evaluated the placebo-subtracted difference in means of percentage triglyceride change at week 24. Mixed-model repeated measures were used for statistical modeling.

RESULTS Of 229 patients, 226 (mean [SD] age, 55 [11] years; 176 male [78%]) were included in the primary analysis. Baseline mean (SD) triglyceride level was 897 (625) mg/dL and plasma APOC3 level was 32 (16) mg/dL. Plozasiran induced significant dose-dependent placebo-adjusted least squares (LS)-mean reductions in triglyceride levels (primary end point) of -57% (95% CI, -71.9% to -42.1%; $P < .001$), driven by placebo-adjusted reductions in APOC3 of -77% (95% CI, -89.1% to -65.8%; $P < .001$) at week 24 with the highest dose. Among plozasiran-treated patients, 144 of 159 (90.6%) achieved a triglyceride level of less than 500 mg/dL. Plozasiran was associated with dose-dependent increases in low-density lipoprotein cholesterol (LDL-C) level, which was significant in patients receiving the highest dose (placebo-adjusted LS-mean increase 60% (95% CI, 31%-89%; $P < .001$). However, apolipoprotein B (ApoB) levels did not increase, and non-high-density lipoprotein cholesterol (HDL-C) levels decreased significantly at all doses, with a placebo-adjusted change of -20% at the highest dose. There were also significant durable reductions in remnant cholesterol and ApoB48 as well as increases in HDL-C level through week 48. Adverse event rates were similar in plozasiran-treated patients vs placebo. Serious adverse events were mild to moderate, not considered treatment related, and none led to discontinuation or death.

CONCLUSIONS AND RELEVANCE In this randomized clinical trial of patients with sHTG, plozasiran decreased triglyceride levels, which fell below the 500 mg/dL threshold of acute pancreatitis risk in most participants. Other triglyceride-related lipoprotein parameters improved. An increase in LDL-C level was observed but with no change in ApoB level and a decrease in non-HDL-C level. The safety profile was generally favorable at all doses. Additional studies will be required to determine whether plozasiran favorably modulates the risk of sHTG-associated complications.

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Severe hypertriglyceridemia (sHTG), ie, plasma triglyceride level greater than 500 mg/dL (to convert triglycerides to millimoles per liter, multiply by 0.0113) is typically multifactorial, involving genetic susceptibility and a combination of diet, lifestyle, alcohol intake, drugs, and/or comorbidities such as hypothyroidism, obesity, and type 2 diabetes.^{1,2} sHTG prevalence in the US is approximately 1.7%.^{1,3} Elevated triglyceride levels can increase risk of atherosclerotic cardiovascular disease (ASCVD), nonalcoholic fatty liver, or steatohepatitis, and acute pancreatitis, often with recurrent attacks and associated adverse outcomes.⁴⁻⁷ sHTG is estimated to account for 10% to 20% of acute pancreatitis cases.⁸⁻¹⁰ Acute pancreatitis risk is proportional to the number, characteristics, and concentration of triglyceride-rich lipoproteins (TRLs), particularly chylomicrons, prior acute pancreatitis episodes, and increases linearly for triglyceride levels greater than 500 mg/dL.¹¹

sHTG is treated with diet, exercise, and adjunctive pharmacotherapy if levels persist greater than 500 mg/dL. The US Food and Drug Administration has approved the use of fibrates and ω -3 fatty acids such as icosapent ethyl to treat sHTG. Some patients with sHTG have sustained chylomicronemia (triglycerides >880 mg/dL) and lack bioavailable lipoprotein lipase (LPL), the key TRL hydrolysis enzyme. Patients with LPL deficiency have very high acute pancreatitis risk and do not respond to fibrates or ω -3 fatty acids.¹²⁻¹⁴ Although triglyceride level reduction to less than 500 mg/dL is considered sufficient to prevent recurrent pancreatitis,¹³ this threshold is often difficult to achieve.¹⁴ Available therapeutic agents modestly reduce fasting triglycerides by 10% to 30%, but evidence of reduced acute pancreatitis incidence is lacking.¹⁵

Apolipoprotein C3 (APOC3) is a glycoprotein produced primarily in the liver and intestine, and it is carried by TRLs such as very-low-density lipoprotein (VLDL) and chylomicrons. APOC3 plays a key role in inhibiting LPL, thereby reducing the metabolism of TRLs by adipocytes and slowing hepatic TRL uptake, contributing to both hypertriglyceridemia and chylomicronemia. APOC3 also contributes to TRL management through LPL-independent mechanisms¹⁵ and is involved in TRL plasma clearance. APOC3 loss-of-function gene variants are independently associated with reduced triglycerides and lessened ischemic cardiovascular disease risk.¹⁵⁻¹⁷ In addition, APOC3 loss of function specifically reduces postprandial triglyceride levels, believed to be the key driver of acute pancreatitis in patients with sHTG.¹⁸

Plozasiran is a GalNAc (or N-acetylgalactosamine)-conjugated, hepatocyte-directed, double-stranded small interfering RNA (siRNA) that is administered subcutaneously quarterly and designed to selectively degrade the messenger RNA (mRNA) coding for APOC3. As shown in the phase 1 study data,¹⁹ plozasiran provides significant and durable APOC3 reductions for 12 weeks after a single subcutaneous dose. As a result, it leads to deep and sustained reductions in triglyceride and TRL levels. RNA interference is a catalytic process designed to selectively degrade specific mRNAs. On cellular uptake of the siRNA in hepatocytes, the concentration of intracellular siRNA-loaded RNA-induced silencing complex correlates with targeted mRNA degradation and shows persistence for weeks or months, allowing for infrequent dosing and

Key Points

Question Does plozasiran, an investigational APOC3-targeted small interfering-RNA (siRNA) therapeutic, safely and effectively reduce serum triglyceride level in patients with severe hypertriglyceridemia (sHTG)?

Findings This placebo-controlled, dose-ranging, phase 2b randomized clinical trial of 229 patients with sHTG (fasting triglyceride level ≥ 500 mg/dL) demonstrated that subcutaneously administered plozasiran (dosed on day 1 and week 12) produced significant placebo-subtracted reductions in apolipoprotein C3 level to -77% and mean triglyceride level to -57% at 24 weeks with a significant treatment difference persisting at 48 weeks. Plozasiran demonstrated a generally favorable safety profile.

Meaning This randomized clinical trial demonstrates the potential clinical utility of plozasiran to reduce triglyceride level in patients with sHTG.

an extended pharmacodynamic effect.^{20,21} The rapid clearance of GalNAc-conjugated siRNAs from the bloodstream occurs within 24 to 48 hours after subcutaneous administration.²² This double-blind, placebo-controlled, phase 2b Study to Evaluate ARO-APOC3 in Adults With Severe Hypertriglyceridemia (SHASTA-2) randomized clinical trial was conducted to evaluate the efficacy and safety of repeated administration of plozasiran in patients with sHTG.

Methods

This study was conducted at 74 centers across the US, Europe, New Zealand, Australia, and Canada from May 31, 2021, to August 31, 2023. Study protocol/amendments ([Supplement 1](#); statistical analysis plan available in [Supplement 2](#)) were approved by institutional review boards/independent ethics committees at study sites. All patients provided written informed consent. The study was reported in accordance with the Consolidated Standards of Reporting Trials ([CONSORT](#)) reporting guidelines.

The study included a 6-week screening/diet run-in period and a 3:1 randomization to plozasiran (10, 25, or 50 mg; mechanism of action depicted in eFigure 1 in [Supplement 3](#)) or placebo, stratified by screening triglycerides less than or greater than or equal to 1000 mg/dL with patients allocated unique randomization numbers and randomized by an interactive web response system using a block randomization algorithm. Treatment assignment was blinded (but not dose group as different volumes) with syringes pharmacy-wrapped with translucent wrapping to blind staff and participants (to plozasiran vs placebo). Plozasiran or placebo was given subcutaneously at weeks 0 and 12 with patients followed up to week 48 (eFigure 2 in [Supplement 3](#)), after which they could enroll in an open-label extension. Patients continued a stable diet and concomitant medications per local standard of care (eTable 1 in [Supplement 3](#)).

Adult patients with sHTG (fasting triglyceride levels, 500-4000 mg/dL) were eligible; patients with ASCVD history or elevated risk thereof were required to take lipid-lowering therapy

per local standard of care. Patients with familial chylomicronemia syndrome, pancreatitis within 12 weeks, plasma apheresis within 4 weeks, more than 2-fold elevated liver enzymes, or hemoglobin A_{1c} (HbA_{1c}) level greater than 9% were excluded. Patients self-identified with the following race categories: American Indian or Alaska Native, Asian, Black or African American, Native Hawaiian or Other Pacific Islander, White, unknown, or other (not specified), and patients self-identified with the following ethnicity categories: Hispanic or Latino, not Hispanic or Latino, or not reported. Race and ethnicity information was gathered for the purposes of assessing generalizability of study results.

Statistical Analysis

The primary objective was to evaluate the safety and efficacy of plozasiran in adults with sHTG and to determine a dose for later-stage clinical testing. The primary end point evaluated the difference in means between each dose group and pooled placebo groups of percentage triglyceride change from baseline at week 24 (trough after the second dose); secondary end points included changes in other lipoproteins (APOC3, total apolipoprotein B [ApoB], non-high-density lipoprotein cholesterol [HDL-C], low-density lipoprotein cholesterol [LDL-C], HDL-C, and remnant cholesterol, all adjusted for changes observed in the placebo group). LDL-C level was measured by ultracentrifugation. All samples were taken after a 10-hour (or longer) fast. Exploratory end points included change in liver fat content, acute pancreatitis, and apheresis occurrences. Any inferential statistics (ie, *P* values) for secondary and exploratory end point analyses should be considered exploratory. Safety was analyzed by dose cohort.

Plozasiran dose groups were expected to have 50% to 70% more triglyceride reduction than placebo; with 18 complete per treatment group, the study had greater than 99% power to detect any active treatment dose cohort significantly different from placebo, using a 2-sided test and 1.67% level of significance, adjusted for multiplicity. Considering dropouts and observing for safety, a total of 216 participants were to be randomly assigned to treatment in a 3:1 (active to placebo) ratio within each dose cohort. The primary analysis was performed for all randomized patients who received at least 1 dose of study drug, using a mixed-model repeated-measures approach; adjustment for multiplicity of testing several arms vs placebo was carried out with the Holm step-down procedure. Prespecified subgroups included baseline triglyceride levels of 500 to 2000 mg/dL vs 2000 mg/dL or greater. Missing values were not imputed except for missing/partial adverse event dates, concomitant status for medications and procedures, and for where a complete date was required for calculations. All *P* values were 2-sided, and *P* < .05 was considered statistically significant. Data analyses were performed using SAS software, version 9.4 (SAS Institute).

Results

Study Populations

Baseline characteristics were comparable between treatment groups (Table 1). Of 229 patients, 226 (mean [SD] age, 55 [11]

years; 176 male [78%]; 50 female [22%]) were included in the primary analyses (patient flow diagram in Figure 1). The mean (SD) body mass index was 32 (5; calculated as weight in kilograms divided by height in meters squared), mean (SD) APOC3 level was 32 (16) mg/dL, and the baseline mean (SD) triglyceride and LDL-C levels were 897 (625) mg/dL and 72 (41) mg/dL, respectively. Patients self-identified with the following race categories: 1 American Indian or Alaska Native (0.4%), 9 Asian (4%), 7 Black or African American (3%), 2 Native Hawaiian or Other Pacific Islander (0.9%), 203 White (90%), 1 unknown (0.4%), and 3 other (1%), and patients self-identified with the following ethnicity categories: 33 Hispanic or Latino (15%), 192 not Hispanic or Latino (85%), and 1 not reported (0.4%). A total of 144 patients (64%) had diabetes, and 154 (68%) were receiving statins. In addition, 109 patients (48%) were taking fibrates, 24 (11%) were taking ω-3 fatty acids, and 17 (8%) were taking icosapent ethyl. Thirty-eight of 226 patients (17%) had a history of acute pancreatitis, and 142 (63%) had a history of ASCVD. A full pharmacokinetic analysis was carried out in 48 patients (12 placebo, 36 plozasiran).

Primary and Key Secondary End Points

Results are presented as least squares mean (SE) in Table 2. Treatment with plozasiran was accompanied by dose-dependent placebo-adjusted reductions in triglycerides (primary end point) of −49% (95% CI, −64.0% to −33.7%; *P* < .001), −53% (95% CI, −68.1% to −38.0%; *P* < .001), and −57% (95% CI, −71.9% to −42.1%; *P* < .001), driven by placebo-adjusted reductions in APOC3 of −68% (95% CI, −79.6% to −55.8%; *P* < .001), −72% (95% CI, −83.2% to −59.7%; *P* < .001), and −77% (95% CI, −89.1% to −65.8%; *P* < .001) at week 24, with 10-, 25-, and 50-mg doses, respectively (Table 2 and Figure 2). Significant and durable dose-dependent reductions in APOC3 level persisted through week 48 (36 weeks after the last dose). Significant and durable triglyceride level reductions began as early as week 4 and persisted to week 48 (Figure 2 and Table 2). There was limited interpatient variability in pharmacodynamic response in all plozasiran treatment groups (eTable 2, eTable 3, and eFigure 3 in Supplement 3). Among patients treated with plozasiran, 144 of 159 patients (90.6%) achieved a triglyceride level of less than 500 mg/dL at week 24 and maintained these reductions, with 117 of 153 treated patients (76.5%) remaining below this threshold at week 48. In addition, 77 of 159 patients (48.4%) achieved normal triglyceride levels less than 150 mg/dL at 24 weeks, including 28 of 55 patients (51%) at the highest dose tested (eFigures 3 and 4 in Supplement 3).

Other Secondary and Exploratory End Points

Other lipid parameters were assessed (Table 2 and eTable 2 in Supplement 3). Significant reductions were seen at 24 weeks in non-HDL-C, remnant cholesterol, ApoB48, and APOC2 levels, with no meaningful reductions in lipoprotein (a). Non-HDL-C levels decreased significantly at all doses, with a placebo adjusted change of −20% (95% CI, −30.6% to −9.7%; *P* < .05) at the highest dose; ApoB levels did not increase. Non-HDL-C level reductions remained significant at 48 weeks. Compared with placebo, plozasiran showed substantial and durable dose-dependent reductions as early as 4 weeks in remnant

Table 1. Baseline Characteristics^a

Characteristic	Pooled placebo (n = 60)	Plozasiran			Total (n = 226)
		10 mg (n = 54)	25 mg (n = 55)	50 mg (n = 57)	
Mean (SD) age, y	56 (11)	53 (10)	56 (11)	54 (11)	55 (11)
Sex at birth, No. (%)					
Male	46 (77)	46 (85)	43 (78)	41 (72)	176 (78)
Female	14 (23)	8 (15)	12 (22)	16 (28)	50 (22)
Race, No. (%)					
American Indian or Alaska Native	1 (2)	0	0	0	1 (0.4)
Asian	3 (5)	1 (2)	2 (4)	3 (5)	9 (4)
Black or African American	1 (2)	2 (4)	3 (6)	1 (2)	7 (3)
Native Hawaiian or Other Pacific Islander	0	0	2 (4)	0	2 (0.9)
White	55 (92)	47 (87)	48 (87)	53 (93)	203 (90)
Unknown	0	1 (2)	0	0	1 (0.4)
Other ^b	0	3 (6)	0	0	3 (1)
Ethnicity, No. (%)					
Hispanic or Latino	13 (22)	7 (13)	7 (13)	6 (11)	33 (15)
Not Hispanic or Latino	46 (77)	47 (87)	48 (87)	51 (90)	192 (85)
Not reported	1 (2)	0	0	0	1 (0.4)
BMI, mean (SD) ^c	31 (4)	33 (5)	32 (5)	32 (5)	32 (5)
Triglyceride, mean (SD), mg/dL	851 (507)	890 (577)	942 (756)	908 (653)	897 (625)
Triglyceride, median (IQR), mg/dL	679 (540-929)	696 (559-1088)	598 (517-982)	663 (531-1028)	660 (537-1007)
APOC3, mean (SD), mg/dL ^d	31 (16)	33 (15)	34 (17)	32 (16)	32 (16)
Non-HDL-C, mean (SD), mg/dL	185 (79)	209 (74)	206 (91)	196 (88)	199 (83)
ApoB, mean (SD), mg/dL	95 (29)	103 (44)	103 (32)	110 (55)	102 (41)
Remnant cholesterol UC, mean (SD), mg/dL ^e	115 (82)	134 (88)	132 (98)	124 (92)	126 (90)
LDL-C UC, mean (SD), mg/dL	69 (39)	75 (44)	74 (40)	72 (42)	72 (41)
HDL-C, mean (SD), mg/dL	30 (12)	28 (9)	30 (11)	31 (13)	30 (11)
Receiving statins, No. (%)	41 (67)	35 (65)	39 (71)	39 (70)	154 (68)
High-intensity statins, No. (%)	34 (57)	28 (52)	27 (49)	30 (53)	119 (53)
Fibrates, No. (%)	31 (51)	26 (48)	24 (44)	28 (50)	109 (48)
ω-3 Fatty acids, No. (%)	3 (5)	8 (15)	5 (9)	8 (14)	24 (11)
Icosapent ethyl, No. (%)	4 (7)	3 (6)	5 (9)	5 (9)	17 (8)
Receiving PCSK9i, No. (%)	3 (5)	1 (2)	2 (4)	1 (2)	7 (3)
Receiving GLP1Ra, No. (%)	8 (13)	7 (13)	7 (13)	3 (5)	25 (11)
Diabetes, No. (%)	39 (65)	31 (57)	37 (67)	37 (65)	144 (64)
ASCVD history or at elevated risk for ASCVD, No. (%)	36 (60)	31 (57)	36 (66)	39 (68)	142 (63)
History of acute pancreatitis, No. (%)	13 (22)	6 (11)	8 (15)	11 (19)	38 (17)

Abbreviations: ApoB, apolipoprotein B; APOC3, apolipoprotein C3; ASCVD, atherosclerotic cardiovascular disease; BMI, body mass index; GLP1Ra, glucagon-like peptide 1; HDL-C, high density lipoprotein cholesterol; LDL-C, low density lipoprotein cholesterol; PCSK9i, proprotein convertase subtilisin/kexin type 9; UC, ultracentrifugation.

SI conversion factor: To convert ApoB to grams per liter, multiply by 0.01; to convert cholesterol, HDL-C, and LDL-C to millimoles per liter, multiply by 0.0259; to convert triglycerides to millimoles per liter, multiply by 0.0113.

^a Data are shown for the full analysis set of 226, ie all randomized patients who received at least 1 dose of investigational product.

^b Other race not specified.

^c Calculated as weight in kilograms divided by height in meters squared.

^d Analysis that removed 2 participants with baseline values below limits of quantitation (ad hoc).

^e Based on calculation: total cholesterol - HDL-C - LDL-C (UC).

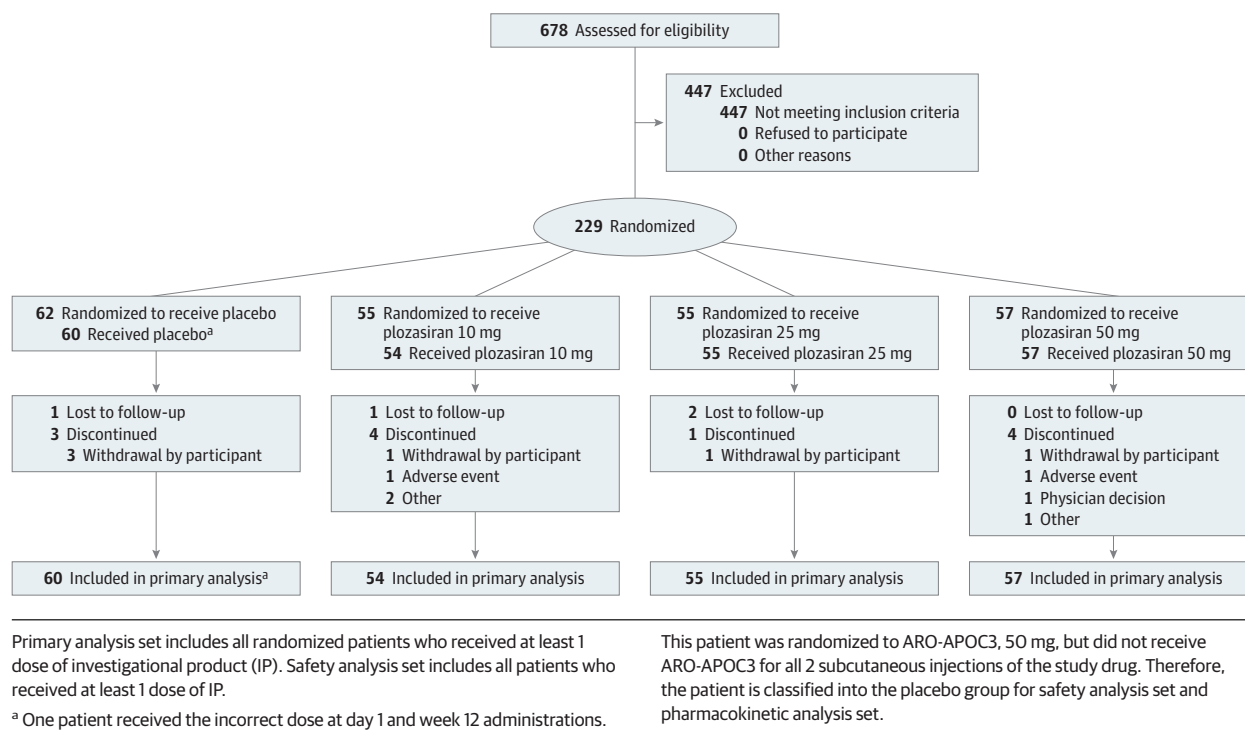
cholesterol level, with reductions of -62% (95% CI, -79% to -44%; $P < .001$), -64% (95% CI, -81% to -46%; $P < .001$), and -59% (95% CI, -76% to -42%; $P < .001$) at week 24 with 10-, 25-, and 50-mg doses, respectively, and maintained significant reductions to week 48 (Table 2 and Figure 2). ApoB48 also showed substantial reductions of -128% (95% CI, -191% to -66%; $P < .001$), -140% (95% CI, -202% to -78%; $P < .001$), -139% (95% CI, -200% to -77%; $P < .001$) at 24 weeks with 10-, 25-, and 50-mg doses, respectively, and maintained significance at 48 weeks (eTable 2 in Supplement 3). Increases were seen in HDL-C, APOA1, and LDL-C levels, with the latter occurring in the context of no increase in ApoB (Table 2 and Figure 2 and eFigure 3 in Supplement 3). Dose-dependent and significant increases in HDL-C level were observed at week 24

and remained significant to week 48. Plozasiran treatment was associated with dose-dependent increases in LDL-C level, which peaked after the second dose. In those receiving the highest dose (50 mg), the placebo-adjusted increase in LDL-C was 60% (95% CI, 31%-89%; $P < .001$); LDL-C steadily declined after 24 weeks and was not significantly different from placebo values at week 48 (Figure 2 and Table 2). Liver fat content (LFC) as assessed by magnetic resonance imaging proton density fat fraction in a subset of patients with LFC greater than 8% at baseline did not change significantly over time (Table 3).

Efficacy Across Subgroups

Changes in triglyceride levels according to tertiles of baseline triglyceride levels are presented in eTable 3 in Supplement 3.

Figure 1. Patient Flow in Study: Consolidated Standards of Reporting Trials (CONSORT)



At 24 weeks, the percentage reductions in triglyceride levels were similar across tertiles, and the mean triglyceride levels reached were well below the threshold associated with elevated acute pancreatitis risk (>500 mg/dL) (eTables 3 and 4 and eFigures 3 and 4 in Supplement 3).

Safety and Tolerability

The adverse event and serious adverse event profiles were similar across treatment groups (eTable 5 in Supplement 3). Observed adverse events generally reflected the comorbidities and underlying conditions of this population. The most common events occurring in 5 or more patients were COVID-19 infection, worsening glycemic control, diarrhea, urinary tract infection, and headache. Worsening glycemic control occurred only in patients with diabetes at baseline. No changes were noted in the Homeostatic Model Assessment for Insulin Resistance (HOMA-IR) test. There was an imbalance in HbA_{1c} level change in the highest-dose group (Table 3 and eFigure 5 in Supplement 3). One patient (50-mg plozasiran group) discontinued study drug due to worsening glycemic control. All serious adverse events were mild to moderate in severity (grade 1-3), except for 1 case of unrelated grade 4 squamous cell carcinoma in a patient in the placebo group, and none led to discontinuation or death. Acute pancreatitis was adjudicated by an independent committee according to the revised Atlanta criteria.²³ Three acute pancreatitis episodes (Table 3) occurred during the randomized period in 2 of 61 patients (3.3%) in the placebo group, and 1 episode occurred in 1 of 165 patients (0.6%) receiving plozasiran, 50 mg (odds ratio, 0.18; 95% CI, 0.02-2.02). All patients who had acute pancreatitis in this

study had a previous history of acute pancreatitis. The event in the patient assigned to the 50-mg plozasiran cohort occurred during the safety observation period 9 months after the final dose, at which time the patient's triglyceride levels had returned to baseline level of greater than 2000 mg/dL from a treatment-period nadir of 106 mg/dL. All episodes of acute pancreatitis resolved.

Discussion

As severely elevated serum triglyceride levels are an established risk factor for acute pancreatitis and ASCVD, more effective triglyceride-lowering therapies are desired to improve patient outcomes.^{5,8,11} The findings in this randomized clinical trial demonstrate that reducing APOC3 with plozasiran markedly reduced triglyceride concentrations in a dose-dependent manner in patients with sHTG and appeared to be well tolerated with an acceptable safety profile. At higher doses, plozasiran therapy reduced the triglyceride concentration to a substantial degree when adjusted for placebo as compared with currently available drugs. Nearly all (>90%) patients who received plozasiran achieved a triglyceride concentration of less than 500 mg/dL at 24 weeks, the generally quoted risk threshold for acute pancreatitis. These effects were consistent across key subgroups of interest with minimal interpatient variability in pharmacodynamic response. The persistent efficacy of plozasiran was unsurprising and accounts for the 12-week dosing interval selected, consistent with its recognized mechanism as an RNA interference therapeutic.

Table 2. Percentage Change From Baseline for End Points at Weeks 24 and 48^a

Serum concentration	Placebo (n = 60)	Plozasiran		
		10 mg (n = 54)	25 mg (n = 55)	50 mg (n = 57)
Primary and key secondary and exploratory end points				
TG ^b				
Baseline, mean (SD), mg/dL	851.1 (507.0)	889.5 (577.1)	941.8 (756.2)	907.9 (653.4)
Absolute change at week 24, mean (SD), mg/dL	−141.31 (682.54)	−560.60 (402.00)	−711.30 (791.19)	−672.15 (510.33)
LS % change at week 24, mean (SE)	−17.2 (5.3)	−66.0 (5.6)	−70.2 (5.5)	−74.2 (5.4)
LS mean of difference vs placebo (95% CI)	NA	−48.8 (−64.0 to −33.7) ^c	−53.1 (−68.1 to −38.0) ^c	−57.0 (−71.9 to −42.1) ^c
LS % change at week 48, mean (SE)	−6.6 (7.7)	−31.4 (8.2)	−58.0 (8.2)	−53.4 (8.0)
LS mean of difference vs placebo (95% CI)	NA	−24.9 (−47.1 to −2.7) ^d	−51.5 (−73.6 to −29.3) ^c	−46.8 (−68.7 to −25.0) ^c
Baseline, median (IQR), mg/dL	678.6 (539.6 to 929.1)	696.0 (558.8 to 1087.6)	597.7 (517.3 to 982.1)	663.1 (530.8 to 1028.3)
% Change at week 24, median (IQR)	−29.4 (−55.5 to 5.1)	−74.9 (−82.0 to −64.2)	−76.0 (−87.0 to −62.4)	−80.2 (−85.2 to −70.2)
Median of difference vs placebo (95% CI)	NA	−43.0 (−54.1 to −31.9) ^c	−47.5 (−58.6 to −36.4) ^c	−48.4 (−58.0 to −38.7) ^c
APOC3 ^e				
Baseline, mean (SD), mg/dL	30.6 (15.9)	33.4 (15.4)	34.5 (16.8)	31.6 (16.1)
LS % change at week 24, mean (SE)	−0.9 (4.2)	−68.5 (4.4)	−72.3 (4.4)	−78.3 (4.3)
LS mean of difference vs placebo (95% CI)	NA	−67.7 (−79.6 to −55.8) ^c	−71.5 (−83.2 to −59.7) ^c	−77.4 (−89.1 to −65.8) ^c
LS % change at week 48, mean (SE)	4.4 (6.12)	−34.3 (6.45)	−48.1 (6.37)	−46.9 (6.20)
LS mean of difference vs placebo (95% CI)	NA	−38.7 (−56.1 to −21.3) ^c	−52.6 (−69.8 to −35.3) ^c	−51.4 (−68.4 to −34.3) ^c
Non-HDL-C				
Baseline, mean (SD), mg/dL	184.8 (78.7)	208.8 (73.6)	206.3 (91.3)	195.7 (87.6)
LS % change at week 24, mean (SE)	−1.5 (3.8)	−29.4 (4.0)	−28.4 (3.9)	−21.7 (3.8)
LS mean of difference vs placebo, %	NA	−27.9 ^c	−26.9 ^c	−20.2 ^d
LS % change at week 48, mean (SE)	0.5 (4.6)	−10.0 (4.9)	−23.0 (4.8)	−13.5 (4.7)
LS mean of difference vs placebo	NA	−10.5	−23.5 ^d	−14.0 ^d
HDL-C				
Baseline, mean (SD), mg/dL	29.7 (11.6)	28.3 (8.8)	29.5 (11.2)	30.5 (12.6)
LS % change at week 24, mean (SE)	10.6 (5.9)	54.2 (6.3)	62.8 (6.1)	67.6 (6.0)
LS mean of difference vs placebo, %	NA	43.5 ^c	52.2 ^c	57.0 ^c
LS % change at week 48, mean (SE)	6.1 (5.4)	23.5 (5.8)	32.4 (5.7)	37.8 (5.5)
LS mean of difference vs placebo	NA	17.5 ^d	26.3 ^d	31.8 ^c
ApoB				
Baseline, mean (SD), mg/dL	96.1 (29.8)	102.6 (44.0)	101.4 (32.3)	110.1 (54.1)
LS % change at week 24, mean (SE)	8.0 (5.9)	6.0 (6.3)	−5.3 (6.1)	0.7 (6.0)
LS mean of difference vs placebo, %	NA	−2.0	−13.3	−7.3
LS % change at week 48, mean (SE)	5.7 (4.1)	6.4 (4.3)	−0.2 (4.2)	2.4 (4.1)
LS mean of difference vs placebo	NA	0.7	−5.9	−3.3
LDL-C (ultracentrifugation)				
Baseline, mean (SD), mg/dL	69.3 (39.3)	74.6 (44.2)	74.3 (39.5)	71.9 (41.9)
LS % change at week 24, mean (SE)	18.0 (10.5)	49.0 (11.1)	43.7 (10.8)	78.2 (10.6)
LS mean of difference vs placebo, %	NA	31.0 ^d	25.8	60.3 ^c
LS % change at week 48, mean (SE)	21.3 (10.7)	33.5 (11.4)	34.4 (11.1)	44.6 (10.9)
LS mean of difference vs placebo	NA	12.2	13.1	23.2
Remnant cholesterol				
Baseline, mean (SD), mg/dL	115.4 (82.2)	134.1 (87.9)	132.1 (98.3)	123.8 (91.5)
LS % change at week 24, mean (SE)	1.8 (6.2)	−60.1 (6.6)	−61.9 (6.4)	−57.0 (6.3)
LS mean of difference vs placebo, %	NA	−61.8 ^c	−63.6 ^c	−58.8 ^c
LS % change at week 48, mean (SE)	1.8 (8.2)	−24.4 (8.8)	−44.9 (8.6)	−34.2 (8.4)
LS mean of difference vs placebo	NA	−26.2 ^d	−46.8 ^d	−36.0 ^d

Abbreviations: ApoB, apolipoprotein B; APOC3, apolipoprotein C3; BLOQ, below limit of quantitation; HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; LS, least squares; TG, triglyceride.

SI conversion factor: To convert ApoB to grams per liter, multiply by 0.01; to convert cholesterol, HDL-C, and LDL-C to millimoles per liter, multiply by 0.0259; to convert triglycerides to millimoles per liter, multiply by 0.0113.

^a Statistical significance was determined using mixed-model repeat-measures analysis.

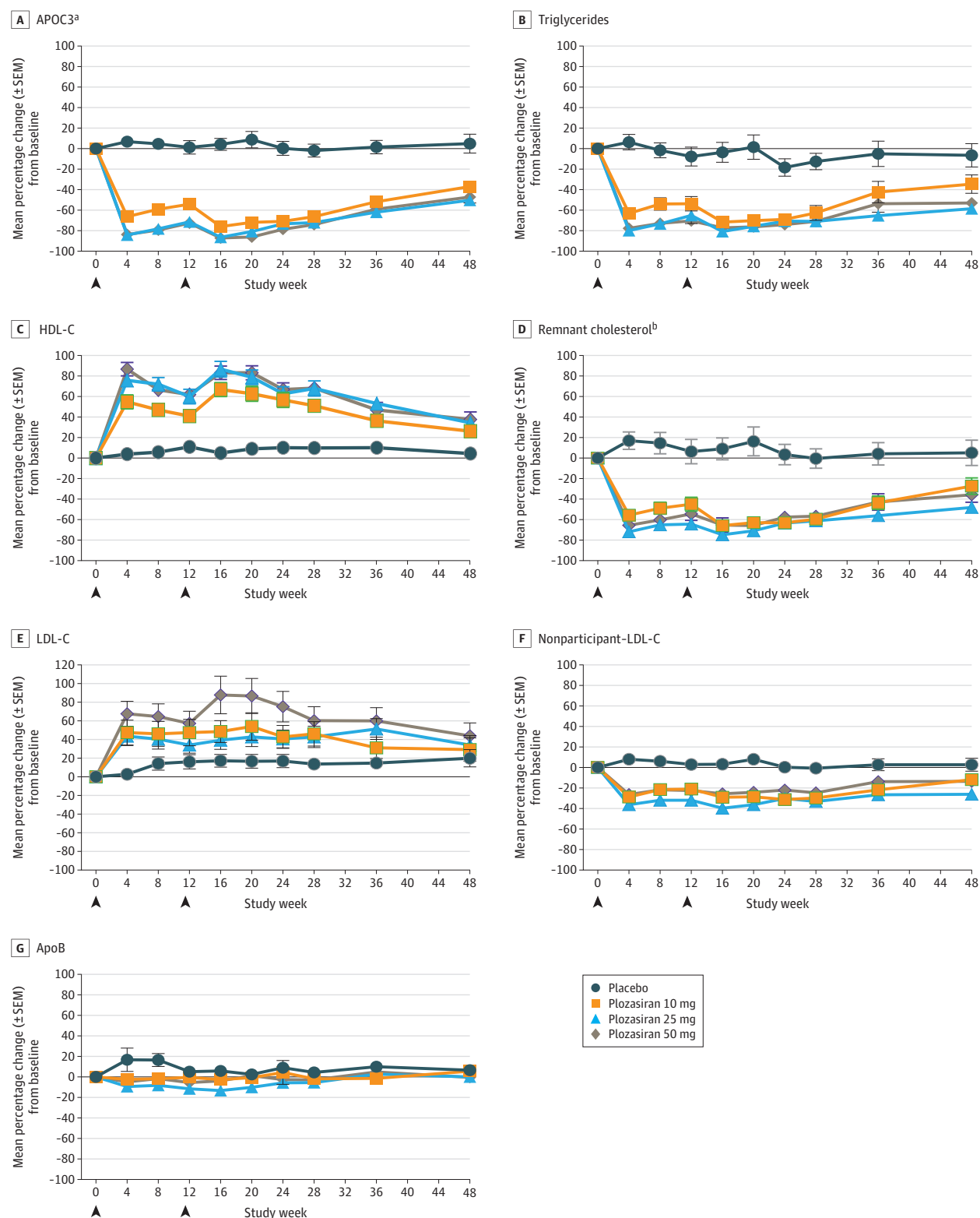
^b Sensitivity analyses assessing the median percentage change in triglyceride levels, using Hodges-Lehmann method with multiple imputation, nonparametric analysis, were generally similar to the primary efficacy analysis.

^c $P < .001$.

^d $P < .05$.

^e Analysis removed 2 participants with BLOQ values at baseline.

Figure 2. Plozasiran Mean Percentage Change of Lipids and Lipoproteins From Baseline Over Time Through Week 48



Data represent mean percentage change ($\pm$ SEM) relative to baseline values. A, Apolipoprotein C3 (APOC3). B, Triglycerides. C, High-density lipoprotein cholesterol (HDL-C). D, Remnant cholesterol. E, Low-density lipoprotein cholesterol (LDL-C), measured by ultracentrifugation. F, Non-HDL-C, based on calculation total cholesterol level - HDL-C level. G, apolipoprotein (ApoB).

Arrowheads indicate the week of injection/administration.

^a Excluding 2 participants with baseline values below limits of quantitation (ad hoc).

^b Based on calculation: total cholesterol - HDL-C - LDL-C level.

Table 3. Key Safety End Points^a

Serum concentration	Placebo (n = 61)	Plozasiran		
		10 mg (n = 54)	25 mg (n = 55)	50 mg (n = 56)
HbA_{1c}^b				
Baseline, mean (SD)	6.5 (1.1)	6.6 (1.3)	6.6 (1.1)	6.5 (1.2)
LS absolute change at week 24, mean (SE)	0.1 (0.1)	0.2 (0.1)	0.1 (0.1)	0.7 (0.1)
LS mean of difference vs placebo (95% CI)	NA	0.1 (−0.2 to 0.4)	0 (−0.3 to 0.3)	0.5 (0.2 to 0.8)
LS absolute change at week 48, mean (SE)	0 (0.1)	0.2 (0.1)	0.1 (0.1)	0.2 (0.1)
LS mean of difference vs placebo (95% CI)	NA	0.2 (−0.1 to 0.5)	0 (−0.3 to 0.3)	0.2 (−0.1 to 0.5)
ALT				
Baseline, mean (SD), U/L	26.3 (9.3)	34.2 (18.4)	33.2 (15.6)	30.3 (17.5)
LS absolute change at week 24, mean (SE)	−3.3 (2.1)	3.5 (2.2)	5.0 (2.1)	6.9 (2.1)
LS mean of difference vs placebo (95% CI)	NA	6.8 (0.9 to 12.8)	8.3 (2.5 to 14.2)	10.2 (4.4 to 16.0)
LS absolute change at week 48, mean (SE)	−4.9 (1.7)	1.2 (1.6)	−1.9 (1.6)	0.9 (1.5)
LS mean of difference vs placebo (95% CI)	NA	6.1 (1.6 to 10.5)	3.0 (−1.4 to 7.4)	5.8 (1.4 to 10.1)
AST				
Baseline, mean (SD), U/L	21.3 (7.8)	24.8 (12.9)	24.2 (9.3)	22.9 (9.8)
LS absolute change at week 24, mean (SE)	−2.7 (1.2)	0.5 (1.3)	2.4 (1.3)	1.6 (1.2)
LS mean of difference vs placebo (95% CI)	NA	3.2 (−0.3 to 6.7)	5.1 (1.7 to 8.5)	4.3 (0.9 to 7.7)
LS absolute change at week 48, mean (SE)	−1.8 (1.2)	1.5 (1.2)	−0.8 (1.2)	0.1 (1.2)
LS mean of difference vs placebo (95% CI)	NA	3.3 (0.1 to 6.5)	1.0 (−2.2 to 4.2)	1.8 (−1.3 to 5.0)
Liver fat (MRI-PDFF), No.^c				
Baseline, mean (SD)	17	9	12	11
LS absolute change at week 24, mean (SE)	17.9 (6.5)	19.3 (5.7)	19.7 (8.8)	22.4 (10.7)
LS mean of difference vs placebo (95% CI)	−0.4 (1.4)	−0.3 (2.3)	0.0 (1.8)	2.3 (1.7)
LS % change at week 24, mean (SE)	NA	0.1 (−5.2 to 5.3)	0.4 (−4.2 to 5.0)	2.7 (−2.0 to 7.3)
LS mean of difference vs placebo, % (95% CI)	0.4 (8.6)	−2.2 (13.7)	1.5 (11.2)	21.3 (10.7)
LS mean of difference vs placebo, % (95% CI)	NA	−2.5 (34.5 to 29.4)	1.1 (−26.9 to 29.2)	20.9 (−7.0 to 48.8)
Acute pancreatitis, adjudicated cases, No. (%)				
Acute pancreatitis	2 (3)	0 (0)	0 (0)	1 (2) ^d
Needing emergent apheresis	0 (0)	0 (0)	0 (0)	0 (0)
Hospitalized	2 (3)	0 (0)	0 (0)	1 (2)

Abbreviations: ALT, alanine transaminase; AST, aspartate aminotransferase; HbA_{1c}, hemoglobin A_{1c}; LS, least squares; MRI-PDFF, magnetic resonance imaging-proton density fat fraction; NA, not applicable.

^a Analysis of covariance model includes covariates treatment arm, stratification factor, and baseline value for liver fat, MRI-PDFF. Statistical significance of HbA_{1c}, ALT, and AST was determined using mixed-model repeat-measures analysis.

^b Patients with diabetes were defined as having HbA_{1c} greater than or equal to 6.5% or fasting glucose greater than or equal to 126 mg/dL or with medical history of diabetes or receiving diabetic medications at baseline.

^c MRI-PDFF analysis set.

^d The event in the patient assigned to the 50-mg plozasiran cohort occurred during the safety observation period c, at which time the patient's triglyceride levels had returned to baseline level of greater than 2000 mg/dL from an on-treatment nadir of 106 mg/dL.

Although the study was not powered to detect a difference in pancreatitis events and the acute pancreatitis incidence was low in this study of short duration, there was a numerically reduced rate with plozasiran-treated patients vs placebo (odds ratio, 0.18; 95% CI, 0.02-2.02). One acute pancreatitis event in a plozasiran-dosed patient occurred after the patient's triglyceride levels had returned to baseline and well outside the expected dosing interval (12 weeks). Plozasiran also decreased ApoB48 level, a marker of chylomicron particle number, as well as remnant cholesterol. These findings support a mechanistic role of plozasiran in reducing acute pancreatitis risk by promoting chylomicron and remnant catabolism through both LPL-dependent and LPL-independent mechanisms.²⁴ These data are consistent with a recent report that similarly suggested that inhibition of *APOC3* expression with the antisense oligonucleotide volanesorsen reduced acute

pancreatitis occurrence in a meta-analysis of patients with HTG and sustained chylomicronemia due to LPL deficiency and familial chylomicronemia syndrome (FCS), sHTG, or familial partial lipodystrophy.²⁵ Taken together, these results support the concept that deep and sustained inhibition of *APOC3* may provide meaningful protection against a potentially fatal condition that currently has no approved effective therapies.

Although these results provide proof of concept that *APOC3* silencing with plozasiran can effectively lower triglyceride levels in patients with sHTG, treatment with plozasiran also led to meaningful and statistically significant reductions in TRLs, which are also thought to be atherogenic.^{26,27} In addition, these changes across the lipid profiles were observed in the setting of significant decreases in *APOC3*, which itself is proinflammatory and has been proposed to be an independent cardiovascular risk factor and a predictor of

major cardiovascular events independent of lipid profiles in some patient populations.²⁸

At baseline, the patients in this trial had mean LDL-C levels ranging from 69 to 75 mg/dL (to convert to millimoles per liter, multiply by 0.0259) across the patient cohorts with 68% taking statin therapy. Increases in LDL-C level were seen in response to plozasiran therapy but were accompanied by no change in ApoB level, the critical structural protein of all atherogenic lipoproteins. LDL-C-level increases are expected in patients with sHTG based on the mechanism of action of plozasiran and are consistent with studies using other APOC3 inhibitors (eg, volanesorsen) after rapid triglyceride reductions. This phenomenon is likely a result of rapid TRL conversion to LDL particles in the setting of removal of APOC3-mediated inhibition of LPL and decreased cholesteryl ester transfer protein activity, which exchanges LDL-C for triglycerides in VLDL particles.²⁹ This indicates that attention should be paid to LDL-C levels when potent triglyceride-lowering therapies are used and LDL-directed therapies either added or modified accordingly. With a growing body of evidence implicating remnant cholesterol as an independent causal risk factor in ASCVD, the physiologic increases in LDL-C level observed here may ultimately be offset by the attendant decreases in triglyceride level, remnant cholesterol level, non-HDL-C level, and lack of increase in ApoB level.³⁰ Although over half of the patients enrolled in this study had a history of ASCVD or were considered high risk for ASCVD, cardiovascular outcomes trials will be needed to evaluate the effect of plozasiran on cardiovascular events.

Although pharmacodynamic response in the plozasiran treatment groups was highly consistent, the variability was slightly higher than that reported with siRNA therapies targeted against other lipid targets in phase 2 trials.³¹ This is likely due to the fact that triglyceride levels as an end point are driven by both genetics and behavioral factors vs monogenetically driven targets like lipoprotein (a). Furthermore, in at least 1 patient treated with 25-mg plozasiran, an apparent lack of treatment effect was later found to be due to elevated free glycerol levels causing pseudohypertriglyceridemia, due to a glycerol kinase deficiency.³² Once the diagnosis was confirmed, this patient's triglyceride value was corrected for free glycerol, and they were also found to have a decrease in triglyceride levels, in keeping with most other treated patients. This patient's corrected data were not included in the dataset reported here (Daniel Gaudet, MD PhD, written personal communication, March 27, 2024).³³

Regarding safety, plozasiran led to a modest worsening of glycemic control relative to placebo in the highest dose group tested, leading to study discontinuation in one patient. In an attempt to reduce additional unplanned interventions during the trial, investigators were instructed to limit changes to baseline medications. Glycemia-related events were confined to patients with diabetes at baseline. HOMA-IR assessments did not suggest a change in insulin sensitivity (eFigure 5 in Supplement 3).

This may point to a change in postprandial glycemia, which could result from increased substrate delivery to the liver to drive gluconeogenesis, due to enhanced hydrolysis of chylomicrons and remnants. This is a hypothesis that can be further explored. Importantly, this signal of worsening glycemic control was also seen in the Efficacy and Safety of Volanesorsen in Patients With Multifactorial Chylomicronaemia (COMPASS) study and similarly was only observed in patients with diabetes at baseline.³⁴ Moreover, recently reported volanesorsen long-term use up to 5 years in patients with FCS has not revealed any chronic deleterious effects on glucose homeostasis.³⁵ This suggests that irrespective of mechanism, the worsening glycemic control observed with APOC3 silencing in sHTG may be transient in nature and/or can be readily mitigated in real-world practice with attention to antidiabetic therapy.³⁶

Limitations

A limitation of our trial is that the treatment consisted of 2 dose administrations, expected to cover a 24-week dosing period, with a 24-week follow-up period to assess safety. Although an extended safety follow-up phase was implemented as part of the current trial, a larger and longer trial is necessary for the complete evaluation of the long-term efficacy and safety of plozasiran. In addition, enrolled patients lacked ethnic diversity, thus limiting generalizability of findings to other ethnic demographic groups. Further, beyond the primary analysis, inferential statistics should be considered exploratory and hypothesis generating. The noted numerical reductions in the rate of acute pancreatitis in patients in the plozasiran vs placebo treatment groups needs to be further explored in an outcomes study designed to be suitably powered and numbered to detect these occurrences and should be interpreted with caution.

Some patients in the placebo group experienced large reductions in triglyceride levels. This is not unexpected given the variability in serum triglycerides that has been observed in other studies of investigational agents in sHTG, plus the tendency of trial participants to adhere more closely to requested dietary restrictions. Nevertheless, this variability in the primary end point may have reduced our ability to detect an even greater treatment effect.³⁷

Conclusions

In this randomized clinical trial of adults with sHTG, treatment with plozasiran led to substantial and durable decreases in triglyceride and TRL levels with a generally favorable safety and tolerability profile in this phase 2 trial of moderate size. Results from this study support further development of plozasiran for the treatment of sHTG and exploration of its effects on important clinical events, such as the reduction of acute pancreatitis. Moreover, the pleiotropic effects of plozasiran on atherogenic lipoproteins thought to promote ASCVD remains to be studied in a larger trial.

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